

SYDE 671 Assignment 1

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I will be presenting my solutions to part 1 and part 2 of assignment 1 here.

Part 1 Understanding Zoom

Section 1: Selfie: The wrong way Vs. The right way

I'm clearly really bad at taking selfies so instead I had my wife take photos of me. First standing close without zoom and then stepping away and zoming in while maintaining the size of my face in the photos.



Without zoom



With zoom

Section 2: Architectural Perspective Suppression

Next I took some photos of our tree in the front yard. It's a very old tree and is always the first to change color in fall on the entire street.



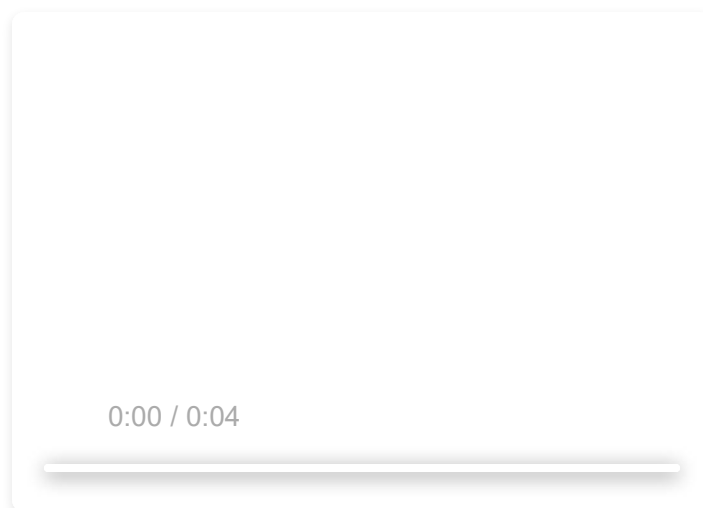
Our tree close-up, without zoom



Our tree from a distance, with zoom

Section 3: Dolly Zoom

For dolly zoom demonstration, I placed a vase and took several photos of it each time moving away and zooming in on it.



Dolly zoom GIF

Part 2: Digitizing Prokudin-Gorskii Glass Plate Images

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Welcome to Part 2 of the assignment. In this phase, I dive deeper into advanced image processing. We will be colorizing scenes from the Russian Empire captured by Sergei Mikhailovich Prokudin-Gorskii, which survive as three vertically stacked grayscale channel exposures (Blue, Green, and Red).

My approach to this assignment was broken into three primary stages:

- **Stage 1: Single-Scale Channel Splitting & Alignment** — Splitting the raw vertically stacked glass plate into separate channels and computing baseline displacements on low-resolution inputs.
- **Stage 2: Multi-Scale Gaussian Pyramid Downsampling** — Constructing coarse-to-fine pyramids to dramatically accelerate alignment search for high-resolution images.
- **Stage 3: Fine-Tuning & Border Crop Optimization** — Offsetting metric evaluation away from plate margins to eliminate border artifacts and sharpen structural detail.

1. Single-Scale Baseline Alignment

For Part 2, the primary goal was to take a single grayscale glass plate image containing vertically stacked exposures captured through blue, green, and red filters, and align any two color channels relative to the third to produce a full-color RGB composite.

I started off with the smaller JPEG images included in the dataset to test the logic and functionality of the program before analyzing the larger high-resolution plates.

The input glass plate was split into separate Blue, Green, and Red channel arrays by slicing the image into three equal sections along its height (each 1/3 of the total height) while keeping full width for all three channels. After isolating the channels, I selected the **Blue channel as the reference frame** to align the Green and Red channels against.

Using a search margin of ± 15 pixels—defining a search window of 961 candidate displacements (31 x 31)—I used the L2 scoring method (SSD) to calculate the optimal (dx, dy) displacement vectors for both Green and Red. Once determined, both channels were shifted and stacked with the Blue channel to produce a colored image. While this single-scale approach was effective for smaller images, it became computationally slow when applied to high-resolution plates.

2. Multi-Scale Gaussian Pyramid Downsampling

To improve efficiency and speed up processing for larger images, I implemented the image pyramid method to downsample the images across 5 scale levels. Downsampling effectively reduced the total pixel count at each stage, drastically lowering the number of calculations required to determine optimal displacement vectors.

While the multi-scale concept is straightforward, I encountered a couple of key technical challenges:

- **Kernel Optimization:** My first implementation used a standard 3 x 3 blurring kernel, which still took over 4 minutes per image. After further testing, I switched to a separable 1 x 5 kernel pass, which significantly accelerated computation. Applying separable 1D horizontal and vertical filtering requires far fewer operations per pixel than evaluating a full 2D matrix at once.
 - **Border Noise & Metric Distortion:** My second mistake was evaluating alignment scores over the full channel area. Because the glass plate channels had uneven, dark, and thick outer borders, the correlation metric was distorted by edge artifacts—leading to severe misalignments and noticeable color fringing.
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3. Fine-Tuning & Margin Crop Optimization

Initially, evaluating alignment scores over the entire frame produced poor results—high-frequency details like train tracks failed to overlap cleanly to form a sharp picture.



Figure 1A: Misaligned result caused by full-frame border artifacts skewing displacement vectors.



Figure 1B: Sharp alignment achieved after introducing a 12% margin crop during scoring.

To solve this, I shifted metric evaluation away from outer margins toward the center of the frame. By introducing a **12% border margin crop** during similarity calculations, the scoring function evaluated only high-intensity interior content. This produced significantly sharper, better-aligned composite images.

While color fringing still appears along the outer borders due to raw plate edges, this can easily be trimmed with a final post-processing crop to keep focus strictly on the internal scene.

4. Results & Comparative Analysis

Below are additional images processed with the refined pipeline, demonstrating varying outcomes based on plate quality and exposure characteristics:



Figure 2: Optimal output. High border brightness and balanced exposure prevented border noise from distorting displacements.

Figure 2 represents probably the best image that the program produced. Since the borders on this plate had high brightness and low exposure variance, there was no severe color fringing caused by misaligned displacements.



Figure 3: Aligned composite exhibiting a heavy blue and green color cast across the sky and grass.

While Figure 3 aligned spatially well, it displays heavy blue and green color shifts across the sky and grass. This could be due to Blue being incorrectly chosen as the static reference channel for this specific exposure, or because color channel

weighting was unbalanced. Implementing automated white balancing post-processing would refine these color dynamics further.



Figure 4: Perfectly aligned and well-balanced reconstruction.

Figure 4 is another good example where the red, blue, and green channels are well aligned to produce a sharp image with very little border distortion or color fringing.



Figure 5: Example of residual color fringing along outer plate borders.

Figure 5 shows strong central subject alignment, but highlights residual color fringing along the outer plate edges where raw frame margins overlap unequally.